Practical · The circumference and area of a circle

Cambridge insert: the one shape with no straight sides, and the number π that measures it.

LESSON 51 | 1 × 40 MIN | GEOMETRY 8, TEMA 52 | STAGE 9 · 7.1

LESSON CLOCK

6' | 12' | 12' | 6' | 4'

Measure a round lid: C ÷ d	6 min
Circumference	12 min
Area	12 min
Arcs and sectors	6 min
Homework	4 min

LEARNING OBJECTIVES

- Use $C = 2\pi r$ and $S = \pi r^2$.
- Find the radius from a given circumference or area.
- Find the area of a sector and the length of an arc.
- Solve practical problems involving circles.

TEXTBOOK

National	Тема 52, pp. 122–125
Cambridge	Learner's Book pp. 142–148
Workbook	Workbook 7.1

01 Explanation

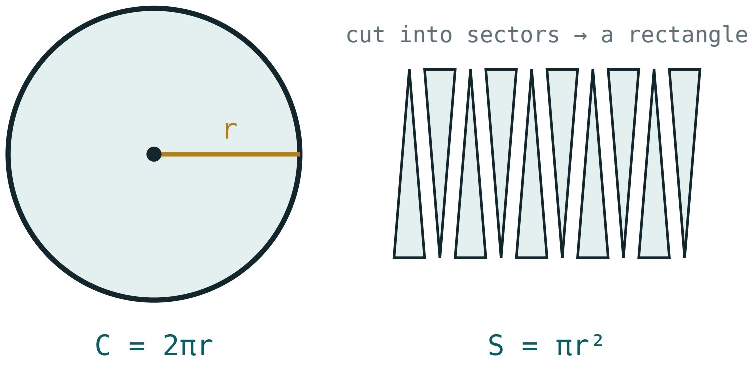
π, and the two formulas

Measure the circumference of any round object and divide by its diameter. Every time, for every circle, you get the same number:

$$\pi = \frac{C}{d} \approx 3.14159\ldots$$

CIRCUMFERENCE AND AREA

$$C = 2\pi r = \pi d \mid S = \pi r^2$$



Cut the circle into thin sectors and interleave them: the result approaches a rectangle of base πr and height r.

Why πr^2? Cut the circle into many thin sectors and lay them alternately up and down. The result is nearly a rectangle of height r and base half the circumference, πr — giving $\pi r \cdot r = \pi r^2$. The thinner the sectors, the better the fit.

RADIUS OR DIAMETER?

$S = \pi r^2$ needs the **radius**. Given a diameter of 10, the area is $\pi \cdot 5^2 = 25\pi$, not 100π — a factor of four out.

Arcs, sectors and rings

A sector of angle α is the fraction $\frac{\alpha}{360°}$ of the whole circle, so both its arc and its area are that same fraction:

$$\ell = \frac{\alpha}{360°} \cdot 2\pi r \;\mid\; S_{sector} = \frac{\alpha}{360°} \cdot \pi r^2$$

A quarter circle of radius 8: arc $\frac{1}{4} \cdot 16\pi = 4\pi \approx 12.6$, area $\frac{1}{4} \cdot 64\pi = 16\pi \approx 50.3$.

A **ring** (annulus) between radii R and r has area $\pi(R^2 - r^2)$ — the big disc less the small one, by additivity.

Give answers exactly in terms of π when possible, and only round at the very end.

02 Worked examples

EXAMPLE 1 **A circle has radius *7 cm*. Find its circumference and area, taking $\pi \approx 3.14$.**

1. $C = 2\pi \cdot 7 = 14\pi$
2. ≈ 43.96 cm
3. $S = \pi \cdot 49 = 49\pi$
4. ≈ 153.86 cm^2

ANSWER

$C \approx 44$ cm, $S \approx 153.9$ cm^2

EXAMPLE 2 **A circular pond has area *78.5 m²*. Find its radius and the length of fence needed round it.**

1. $78.5 = 3.14 \cdot r^2$
2. $r^2 = 25$, $r = 5$ m
3. $C = 2 \cdot 3.14 \cdot 5$
4. $= 31.4$ m

ANSWER

$r = 5$ m, fence 31.4 m

EXAMPLE 3

A circular running track has an inner radius of *30 m* and is *4 m* wide. Find the area of the track surface.

1. $R = 34,\ r = 30$
2. $S = \pi(34^2 - 30^2)$

   The ring formula.
3. $= \pi(1156 - 900) = 256\pi$
4. $\approx 804\ m^2$

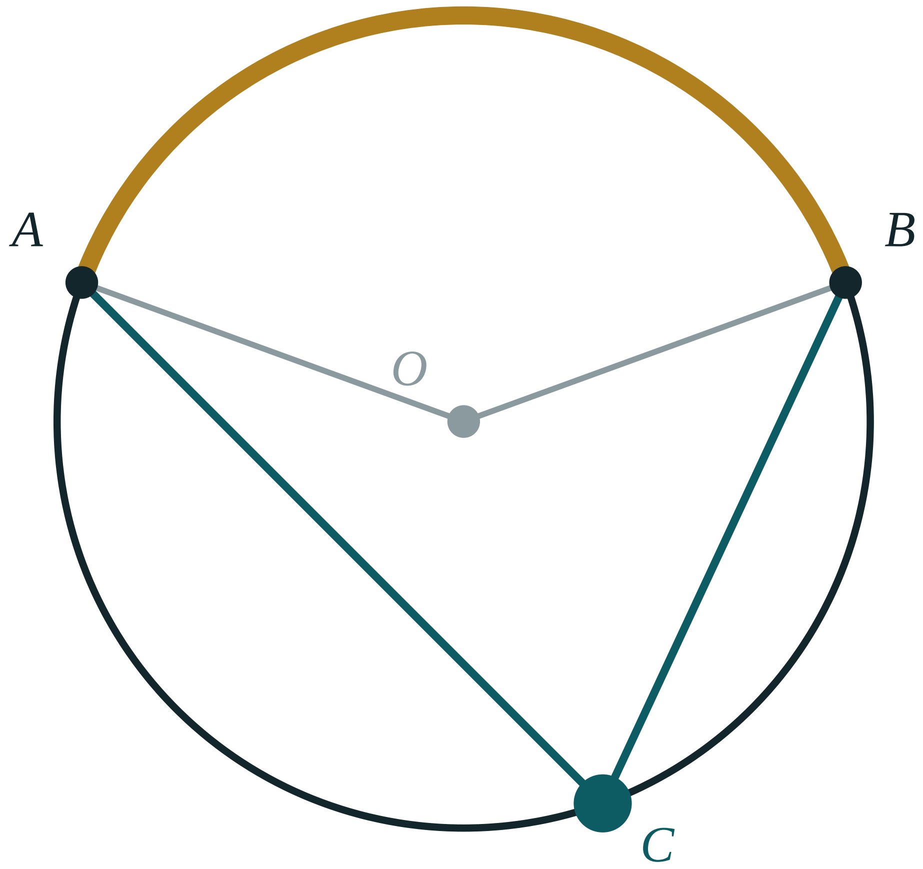
ANSWER

$256\pi \approx 804\ m^2$

03 Interactive model

Explore the circle by dragging the radius; compare the circumference and area readouts.

Around the circle



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Circle	Aylana	Окружность
Circumference	Aylana uzunligi	Длина окружности
Radius	Radius	Радиус
Diameter	Diametr	Диаметр
Pi (π)	Pi soni	Число π
Arc	Yoy	Дуга
Sector	Sektor	Сектор
Semicircle	Yarim aylana	Полуокружность
Annulus (ring)	Halqa	Кольцо

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A circle of radius 5 has circumference:

- 5π
- 10π
- 25π
- 20π

A circle of radius 5 has area:

- 10π
- 25π
- 5π
- 50π

A circle of diameter 12 has area:

- 144π
- 36π
- 12π
- 24π

A quarter circle of radius 4 has area:

- 16π
- 4π
- 8π
- 2π

If the radius doubles, the area:

- doubles
- triples
- quadruples
- stays the same

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find the circumference of a circle of radius 3 (in terms of π).*

   ANSWER 6π

2. *Find the area of a circle of radius 3 (in terms of π).*

   ANSWER 9π

3. *Find the area of a circle of radius 10.*

   ANSWER $100\pi \approx 314$

4. *Find the circumference of a circle of diameter 14.*

   ANSWER $14\pi \approx 44$

5. *Find the area of a semicircle of radius 6.*

   ANSWER 18π

6. *A circle has circumference 20π. Find its radius.*

   ANSWER 10

7. *Find the area of a quarter circle of radius 2.*

   ANSWER π

Medium · the standard question

7 PROBLEMS

1. *A circle has area 49π. Find its radius and circumference.*

   ANSWER $r = 7$, $C = 14\pi$

2. *Find the area of a sector of radius 6 and angle 60°.*

   ANSWER 6π

3. *Find the arc length of a sector of radius 9 and angle 40°.*

   ANSWER 2π

4. *A circular table of diameter 1.2 m is covered. Find the area of cloth needed.*

   ANSWER $0.36\pi \approx 1.13\ m^2$

5. *Find the area of the ring between radii 10 and 6.*

   ANSWER 64π

6. *A wheel of radius 35 cm makes one turn. How far does it travel?*

   ANSWER $70\pi \approx 220$ cm

7. *A circle has circumference 31.4 m. Find its area ($\pi \approx 3.14$).*

   ANSWER $r = 5$, area $78.5\ m^2$

Hard · stretch and reason

7 PROBLEMS

1. *A square of side 10 has a circle inscribed in it. Find the area between them.*

   ANSWER $100 - 25\pi \approx 21.5$

2. *A circle is inscribed in a square of side 10, and a square is inscribed in that circle. Find the smaller square's area.*

   ANSWER 50 — its diagonal is the circle's diameter, 10

3. *A running track is 400 m round with two straights of 100 m and two semicircular ends. Find the radius of the ends.*

   ANSWER $2\pi r = 200$, so $r \approx 31.8$ m

4. *A wheel of diameter 70 cm turns 500 times. How far does it travel, in metres?*

   ANSWER $500 \cdot 0.7\pi \approx 1100$ m

5. *A sector has area 12π and radius 6. Find its angle.*

   ANSWER $120°$

6. *A circular pizza of diameter 40 cm is cut into 8 equal slices. Find the area of one slice.*

   ANSWER $50\pi \approx 157$ cm^2

7. *Two circles have radii in the ratio 2 : 3. Find the ratio of their areas, and explain.*

   ANSWER $4 : 9$ — area depends on the square of the radius, so lengths in ratio k give areas in ratio k^2

07 Homework

Homework — 5 problems

Geometry 8, Тема 52, pp. 122–125. Leave answers in terms of π unless told otherwise.

1. *Find the circumference and area of a circle of radius 8.*
2. *A circle has area 144π. Find its radius and circumference.*
3. *Find the area of a sector of radius 10 and angle 72°.*

4. *Find the area of the ring between radii 12 and 9.*
5. *A bicycle wheel has diameter 66 cm. How far does it travel in 200 turns?*